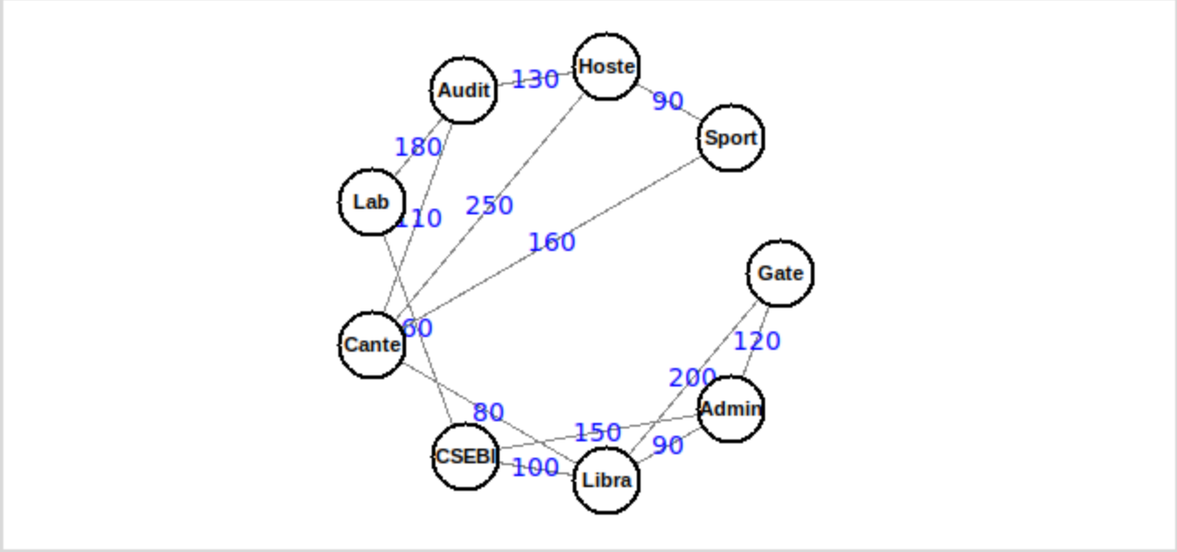
1. Enter paths (Building1 Building2 Distance)

```
Gate Admin 120
Gate Library 200
Admin Library 90
Admin CSEBlock 150
Library CSEBlock 100
Library Canteen 80
CSEBlock Lab 60
Canteen Auditorium 110
Canteen Hostel 250
Auditorium Hostel 130
Lab Auditorium 180
Sports Hostel 90
```

☑ Paths are two-way

Campus paths (sample)

Build | Back | Step | Run all | Reset




**Matrix holds direct paths only. Press Step to begin.**

| | Gate | Admin | Library | CSEBlock | Canteen | Lab | Auditorium | Hostel | Sports |
|---|---|---|---|---|---|---|---|---|---|
| Gate | 0 | 120 | 200 | ∞ | ∞ | ∞ | ∞ | ∞ | ∞ |
| Admin | 120 | 0 | 90 | 150 | ∞ | ∞ | ∞ | ∞ | ∞ |
| Library | 200 | 90 | 0 | 100 | 80 | ∞ | ∞ | ∞ | ∞ |
| CSEBlock | ∞ | 150 | 100 | 0 | ∞ | 60 | ∞ | ∞ | ∞ |
| Canteen | ∞ | ∞ | 80 | ∞ | 0 | ∞ | 110 | 250 | 160 |
| Lab | ∞ | ∞ | ∞ | 60 | ∞ | 0 | 180 | ∞ | ∞ |
| Auditorium | ∞ | ∞ | ∞ | ∞ | 110 | 180 | 0 | 130 | ∞ |
| Hostel | ∞ | ∞ | ∞ | ∞ | 250 | ∞ | 130 | 0 | 90 |
| Sports | ∞ | ∞ | ∞ | ∞ | 160 | ∞ | ∞ | 90 | 0 |

```
Start: direct paths only
```

Find a route

Gate | Sports | Show route

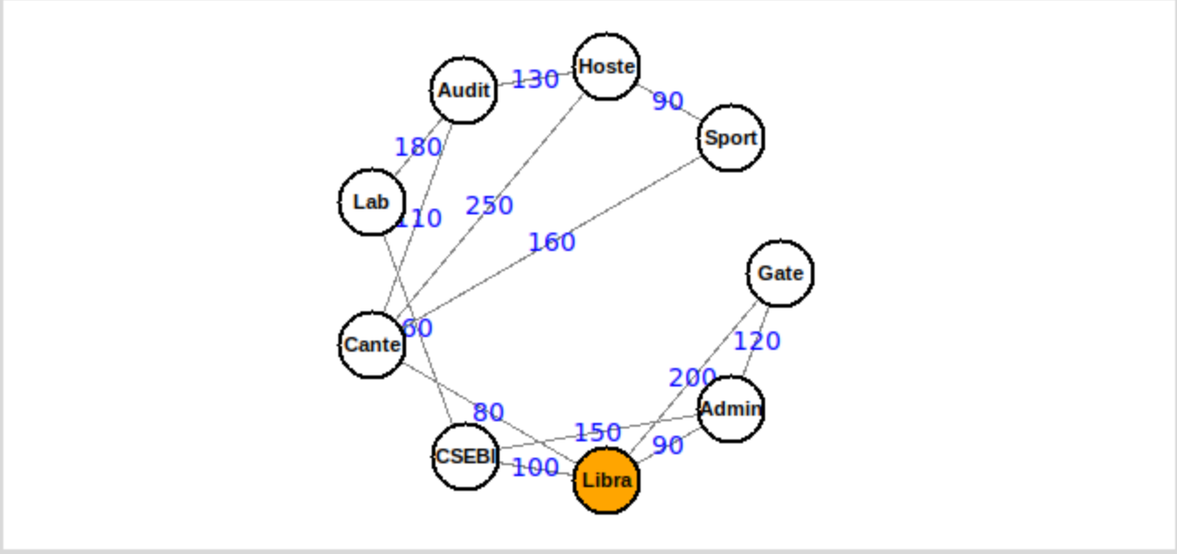
1. Enter paths (Building1 Building2 Distance)

```
Gate Admin 120
Gate Library 200
Admin Library 90
Admin CSEBlock 150
Library CSEBlock 100
Library Canteen 80
CSEBlock Lab 60
Canteen Auditorium 110
Canteen Hostel 250
Auditorium Hostel 130
Lab Auditorium 180
Sports Hostel 90
```

☑ Paths are two-way

Campus paths (sample)

Build | Back | Step | Run all | Reset




**Pivot Library: 6 distance(s) improved.**

| | Gate | Admin | Library | CSEBlock | Canteen | Lab | Auditorium | Hostel | Sports |
|---|---|---|---|---|---|---|---|---|---|
| Gate | 0 | 120 | 200 | 270 | 280 | ∞ | ∞ | ∞ | ∞ |
| Admin | 120 | 0 | 90 | 150 | 170 | ∞ | ∞ | ∞ | ∞ |
| Library | 200 | 90 | 0 | 100 | 80 | ∞ | ∞ | ∞ | ∞ |
| CSEBlock | 270 | 150 | 100 | 0 | 180 | 60 | ∞ | ∞ | ∞ |
| Canteen | 280 | 170 | 80 | 180 | 0 | ∞ | 110 | 250 | 160 |
| Lab | ∞ | ∞ | ∞ | 60 | ∞ | 0 | 180 | ∞ | ∞ |
| Auditorium | ∞ | ∞ | ∞ | ∞ | 110 | 180 | 0 | 130 | ∞ |
| Hostel | ∞ | ∞ | ∞ | ∞ | 250 | ∞ | 130 | 0 | 90 |
| Sports | ∞ | ∞ | ∞ | ∞ | 160 | ∞ | ∞ | 90 | 0 |

```
Start: direct paths only
Pivot Gate: 0 improved
Pivot Admin: 2 improved
  Gate -> CSEBlock: ∞ -> 270
  CSEBlock -> Gate: ∞ -> 270
Pivot Library: 6 improved
  Gate -> Canteen: ∞ -> 280
  Admin -> Canteen: ∞ -> 170
  CSEBlock -> Canteen: ∞ -> 180
  Canteen -> Gate: ∞ -> 280
  Canteen -> Admin: ∞ -> 170
  Canteen -> CSEBlock: ∞ -> 180
```

Find a route

Gate | Sports | Show route

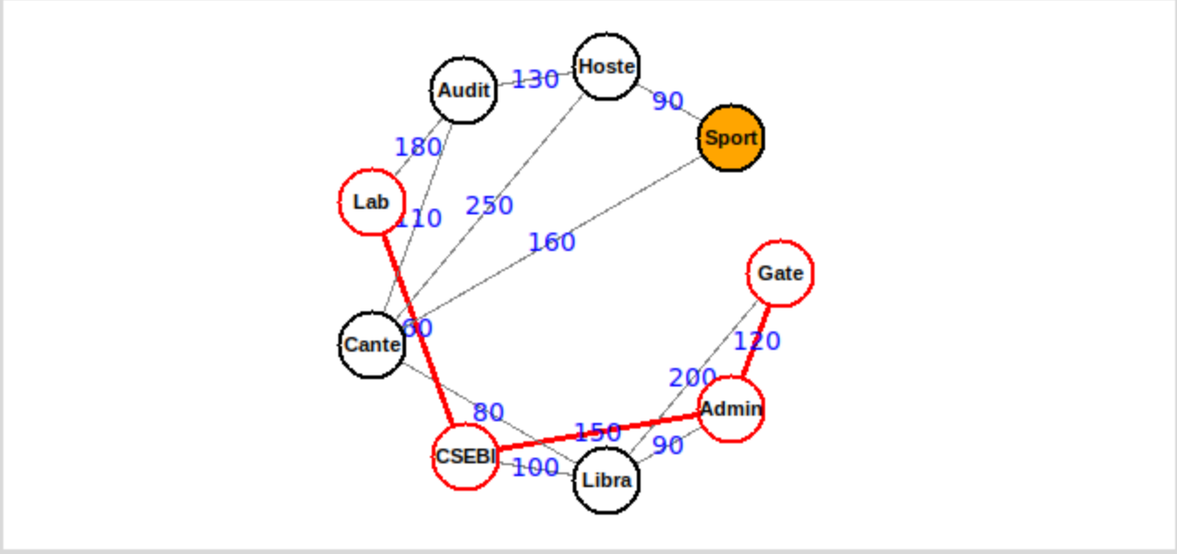
1. Enter paths (Building1 Building2 Distance)

```
Gate Admin 120
Gate Library 200
Admin Library 90
Admin CSEBlock 150
Library CSEBlock 100
Library Canteen 80
CSEBlock Lab 60
Canteen Auditorium 110
Canteen Hostel 250
Auditorium Hostel 130
Lab Auditorium 180
Sports Hostel 90
```

☑ Paths are two-way

Campus paths (sample)

Build | Back | Step | Run all | Reset




**Pivot Sports: 0 distance(s) improved.**

| | Gate | Admin | Library | CSEBlock | Canteen | Lab | Auditorium | Hostel | Sports |
|---|---|---|---|---|---|---|---|---|---|
| Gate | 0 | 120 | 200 | 270 | 280 | 330 | 390 | 520 | 440 |
| Admin | 120 | 0 | 90 | 150 | 170 | 210 | 280 | 410 | 330 |
| Library | 200 | 90 | 0 | 100 | 80 | 160 | 190 | 320 | 240 |
| CSEBlock | 270 | 150 | 100 | 0 | 180 | 60 | 240 | 370 | 340 |
| Canteen | 280 | 170 | 80 | 180 | 0 | 240 | 110 | 240 | 160 |
| Lab | 330 | 210 | 160 | 60 | 240 | 0 | 180 | 310 | 400 |
| Auditorium | 390 | 280 | 190 | 240 | 110 | 180 | 0 | 130 | 220 |
| Hostel | 520 | 410 | 320 | 370 | 240 | 310 | 130 | 0 | 90 |
| Sports | 440 | 330 | 240 | 340 | 160 | 400 | 220 | 90 | 0 |

```
  CSEBlock -> Hostel: 430 -> 370
  Canteen -> Hostel: 250 -> 240
  Lab -> Hostel: 490 -> 310
  Hostel -> Gate: 530 -> 520
  Hostel -> Admin: 420 -> 410
  Hostel -> Library: 330 -> 320
  Hostel -> CSEBlock: 430 -> 370
  Hostel -> Canteen: 250 -> 240
  Hostel -> Lab: 490 -> 310
Pivot Hostel: 2 improved
  Auditorium -> Sports: 270 -> 220
  Sports -> Auditorium: 270 -> 220
Pivot Sports: 0 improved
```

Find a route

Gate | Lab | Show route

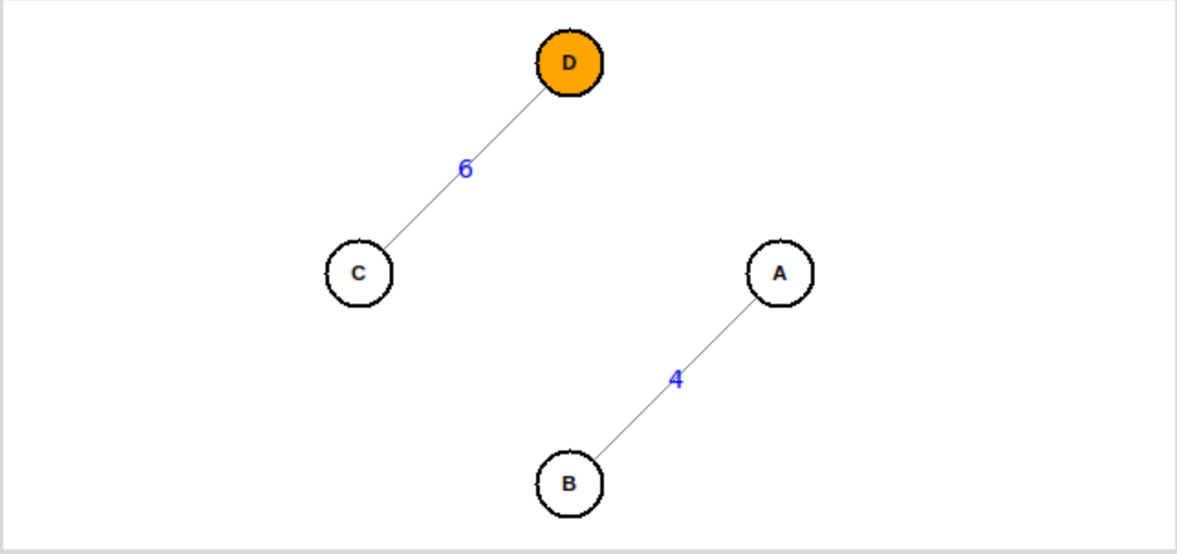
**Gate -> Admin -> CSEBlock -> Lab: 330 m**

```
Best location: Library (1380 m total)

Building        Total   Farthest
Library          1380        320
Canteen          1460        280
CSEBlock         1710        370
Auditorium       1740        390
Admin            1760        410
Lab              1890        400
Sports           2220        440
Hostel           2390        520
Gate             2550        520
```

1. Enter paths (Building1 Building2 Distance)

```
A B 4
C D 6
```




☑ Paths are two-way

Boundary: disconnected buildings

Build | Back | Step | Run all | Reset

**Pivot D: 0 distance(s) improved.**

| | A | B | C | D |
|---|---|---|---|---|
| A | 0 | 4 | ∞ | ∞ |
| B | 4 | 0 | ∞ | ∞ |
| C | ∞ | ∞ | 0 | 6 |
| D | ∞ | ∞ | 6 | 0 |

```
Start: direct paths only
Pivot A: 0 improved
Pivot B: 0 improved
Pivot C: 0 improved
Pivot D: 0 improved
```

Find a route

A | D | Show route

```
No building can reach every other building,
so no best location exists.
```

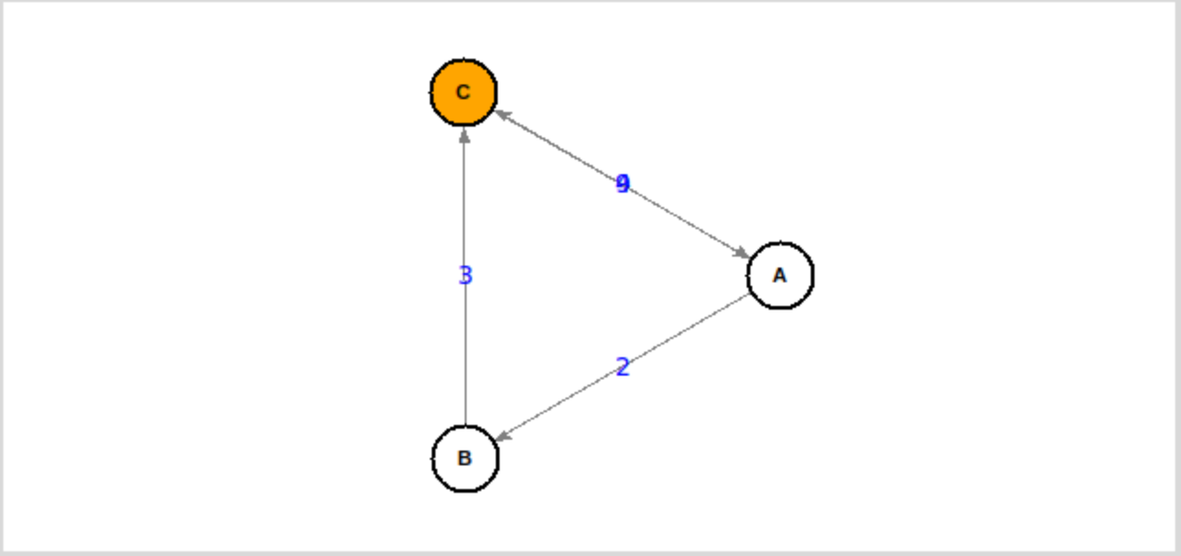
1. Enter paths (Building1 Building2 Distance)

```
A B 2
B C 3
C A 4
A C 9
```

☐ Paths are two-way

One-way paths

Build | Back | Step | Run all | Reset




**Pivot C: 1 distance(s) improved.**

| | A | B | C |
|---|---|---|---|
| A | 0 | 2 | 5 |
| B | 7 | 0 | 3 |
| C | 4 | 6 | 0 |

```
Start: direct paths only
Pivot A: 1 improved
  C -> B: ∞ -> 6
Pivot B: 1 improved
  A -> C: 9 -> 5
Pivot C: 1 improved
  B -> A: ∞ -> 7
```

Find a route

A | C | Show route

```
Best location: A (7 m total)

Building        Total    Farthest
A                   7           5
C                  10           6
B                  10           7
```

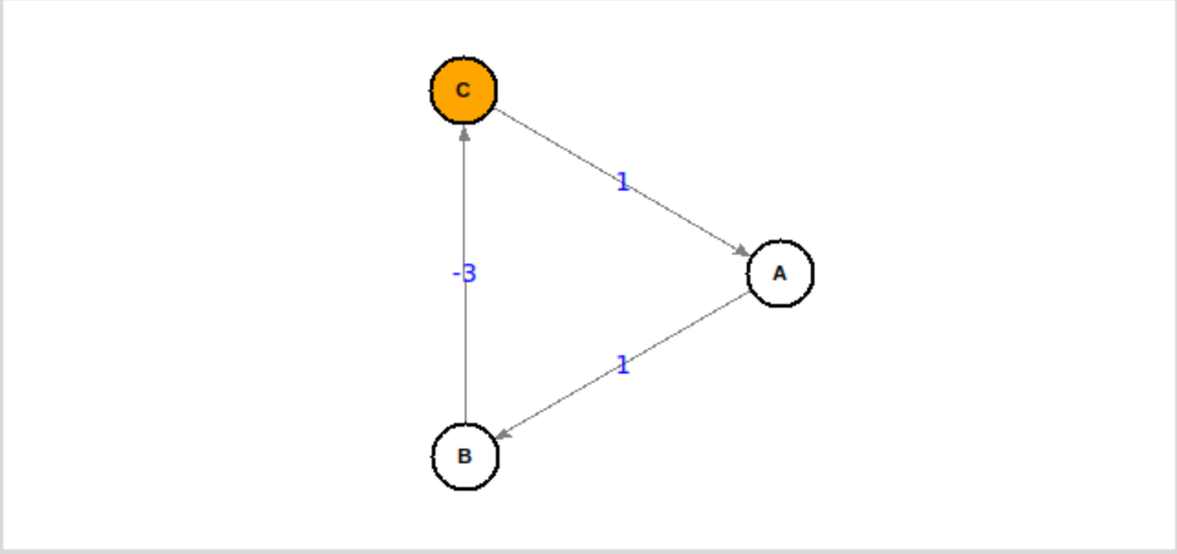
1. Enter paths (Building1 Building2 Distance)

```
A B 1
B C -3
C A 1
```

Paths are two-way

Boundary: negative cycle

Build | Back | Step | Run all | Reset




**Negative cycle detected: a diagonal value is below 0.**

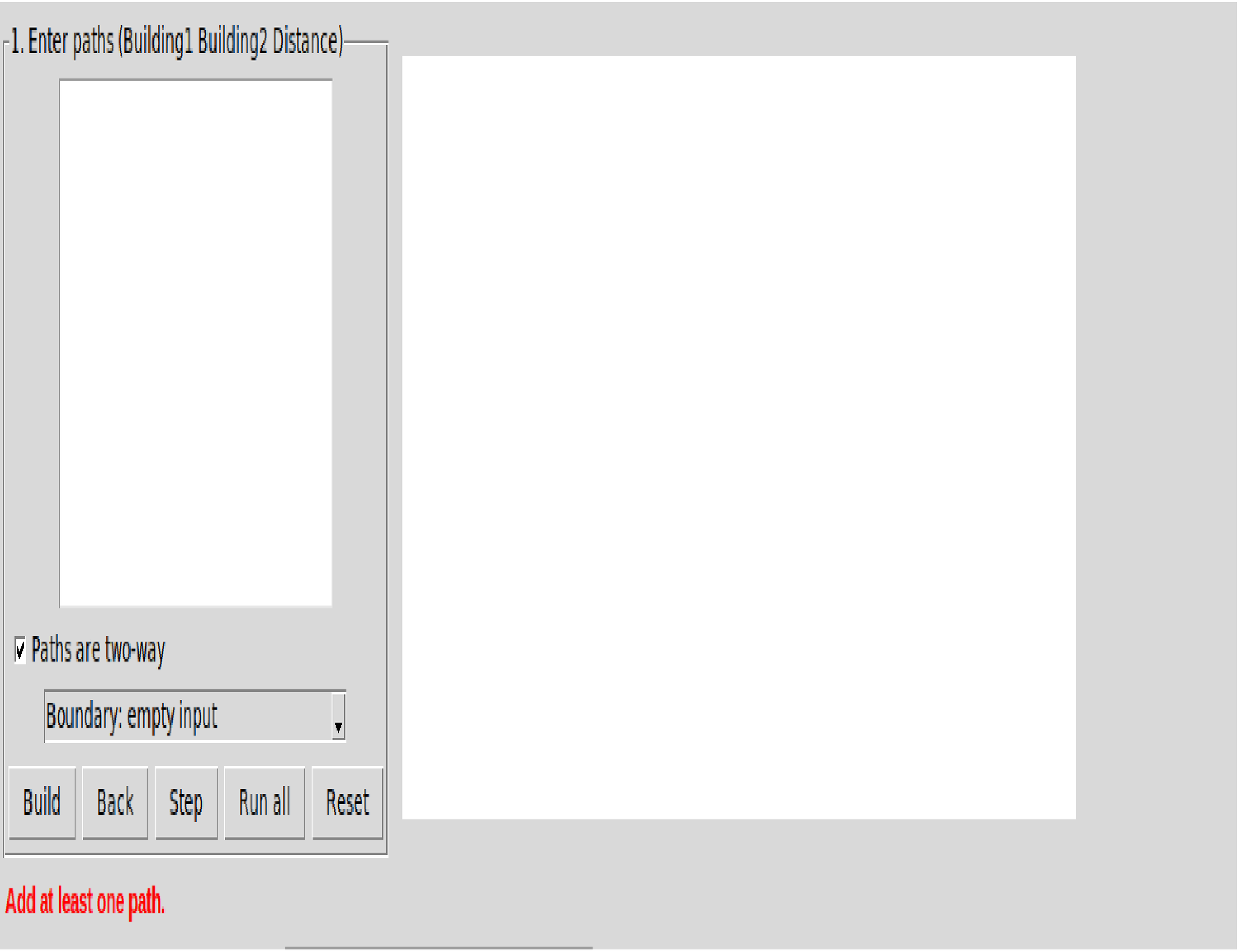
| | A | B | C |
|---|---|---|---|
| A | -1 | 0 | -3 |
| B | -2 | -1 | -4 |
| C | 0 | 1 | -2 |

```
Pivot B: 2 improved
  A -> C: ∞ -> -2
  C -> C: 0 -> -1
Pivot C: 9 improved
  A -> A: 0 -> -1
  A -> B: 1 -> 0
  A -> C: -2 -> -3
  B -> A: ∞ -> -2
  B -> B: 0 -> -1
  B -> C: -3 -> -4
  C -> A: 1 -> 0
  C -> B: 2 -> 1
  C -> C: -1 -> -2
```

Find a route

A | C | Show route

```
Not available: negative cycle.
```


1. Enter paths (Building1 Building2 Distance)
Paths are two-way
Boundary: empty input
Build
Back
Step
Run all
Reset
Add at least one path.

1. Enter paths (Building1 Building2 Distance)

A B x

Paths are two-way

Boundary: invalid distance

Build | Back | Step | Run all | Reset

Line 1: the distance must be a number.